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12.2 Assignment

Compliance and DevOps

After the burst in popularity of DevOps principles in 2010, software companies began abandoning the archaic ways of physical computing infrastructure in favor of modern cloud computing architecture, migrating their monolithic legacy systems to more efficient microservices. However, certain adjustments within the industry must be made to accommodate these new technologies and support the growth of DevOps principles. When the infrastructure of an application changes, its production environments and procedures must change alongside it. Chapter 26 of The DevOps Handbook discusses numerous examples of companies shifting processes in their operations procedures to align with the emergence of DevOps. One of these examples describes a struggle between an enterprise and its compliance auditors, while the other discusses how DevOps strategies can be employed in production to ensure effective and efficient compliance issue detection and correction.

In 2015, Bill Shinn, a security solutions architect with Amazon, noticed a stark contrast in innovations in development strategies and compliance auditing standards. While applications moved to cloud-based systems with auto-scaling capabilities, the compliance auditing processes for those applications lagged behind, still requiring lengthy requests and review procedures and physical proof of compliance in the form of screenshots or logs. To help auditors catch up, he decided that the best way for auditors to do their work efficiently and effectively was to allow them to do so independently by accessing the application's telemetry systems. This way, auditors could request the proof of compliance they needed directly from the system. As a result, the

DevOps Audit Defense toolkit was created to help auditors and development teams understand how this new approach to compliance auditing works.

Backtracking to 2013, a DevOps leader for a financial company made a similar observation about the relationship between compliance auditors and development teams, noticing that auditors still relied on code reviews to detect fraud. Instead, she suggested that they should utilize automated tools to increase the reliability and efficiency of audits. Her proposition worked, and once the change was implemented, the company was able to catch and fix a backdoor planted by one of the company's developers, thanks to the monitoring system they'd implemented. However, perhaps the most important thing to note in favor of this leader's strategy is that the team corrected the issue before the cash audit was due, allowing them to remain in good standing with compliance regulations.

Conclusively, the founding principle of DevOps is decreasing the separation between development and operations, the "Dev" and the "Ops". That said, if the goal is to streamline the relationship between these two teams, it is crucial to allow them to work together while maintaining autonomy. While this idea seems to contradict itself, this goal is perfectly attainable thanks to automated monitoring and telemetry systems. Allowing the teams responsible for operations processes, such as compliance monitoring, to access the information they need efficiently. In the case of the financial company in 2013, this change not only allowed them to streamline the auditing process but also helped them catch and correct a fraud incident early. Once again, proving that following DevOps principles can benefit the entire development team more than maintaining a separation between teams throughout the value stream.

Reference

Kim, G., Debois, P., Willis, J., Humble, J., & Allspaw, J. (2017). The DevOps handbook : how to create world-class agility, reliability, and security in technology organizations. It Revolution Press, Llc.